281. If excess of NH_4OH is added to $CuSO_4$ solution, it forms blue coloured complex which is

(a) $Cu(NH_3)_4SO_4$

(b) $Cu(NH_3)_2SO_4$

(c) $Cu(NH_4)_4SO_4$

(d) $Cu(NH_4)_2SO_4$

Answer: A — (a) $Cu(NH_3)_4SO_4$

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in $Cu(NH_3)_4SO_4$ has the characteristic oxidation state/ligand field that produces the observed colour.

282. Which of the following metals displaces SO_2 gas from concentrated sulphuric acid

(a) Mg

(b) Zn

(c) Cu

(d) None of these

Answer: C — (c) Cu

Chemical reason: The correct choice is Cu. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

283. The method of zone refining of metals is based on the principle of

(a) Greater solubility of the impurity in the molten state than in the solid

(b) Greater mobility of the pure metal than that of the impurity

(c) Higher melting point of the impurity than that of the pure metal

(d) Greater noble character of the solid metal than that of the impurity

Answer: A — (a) Greater solubility of the impurity in the molten state than in the solid

Chemical reason: Zone refining works because impurities are more soluble in the molten metal than in the solid. A moving molten zone carries the impurities toward one end of the bar.

284. A metal when left exposed to the atmosphere for some time becomes coated with green basic carbonate. The metal in question is

(a) Copper

(b) Nickel

(c) Silver

(d) Zinc

Answer: A — (a) Copper

Chemical reason: Copper exposed to moist air and CO_2 develops a green patina of basic copper carbonate, commonly represented as $CuCO_3{\cdot}Cu(OH)_2$.

285. When $CuSO_4$ solution is added toK4[Fe(CN)$_6$], the formula of the product formed is

(a) $Cu_2Fe(CN)_6$

(b) KCN

(c) $Cu(CN)_3$

(d) $Cu(CN)_2$

Answer: A — (a) $Cu_2Fe(CN)_6$

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to $Cu_2Fe(CN)_6$. The choice is fixed by the known composition of that compound/mineral.

286. MnO^{4-} on reduction in acidic medium forms

(a) MnO_2

(b) $Mn+^{+}$

(c) MnO_4-^{-}

(d) Mn

Answer: B — (b) $Mn+^{+}$

Chemical reason: The correct choice is $Mn+^{+}$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

287. Which of the following metals will not react with a solution of $CuSO_4$

(a) Fe

(b) Zn

(c) Mg

(d) Hg

Answer: D — (d) Hg

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Hg. The choice is consistent with the expected oxidation state and stability of the product.

288. Which one of the following metals will not reduce H_2O

(a) Ca

(b) Fe

(c) Cu

(d) Li

Answer: C — (c) Cu

Chemical reason: The correct choice is Cu. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

289. The reaction, which forms nitric oxide, is

(a) C and N_2O

(b) Cu and N_2O

(c) Na and NH_3

(d) Cu and HNO_3

Answer: D — (d) Cu and HNO_3

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Cu and HNO_3. The choice is consistent with the expected oxidation state and stability of the product.

290. A cuprous ore among the following is

(a) Cuprite

(b) Malachite

(c) Chalcopyrites

(d) Azurite

Answer: A — (a) Cuprite

Chemical reason: Cuprite is Cu_2O, an ore containing copper in the +1 (cuprous) state. Malachite and azurite contain Cu(II), and chalcopyrite is a mixed sulfide.

291. When metallic copper comes in contact with moisture, a green powdery/ pasty coating can be seen over it. This is chemically known as

(a) Copper sulphide - Copper carbonate

(b) Copper carbonate - Copper sulphate

(c) Copper carbonate - Copper hydroxide

(d) Copper Sulphate - Copper sulphide

Answer: C — (c) Copper carbonate - Copper hydroxide

Chemical reason: The green patina on copper is basic copper carbonate, usually written $CuCO_3 \cdot Cu(OH)_2$. It forms slowly from Cu in the presence of O_2, H_2O, and CO_2.

292. Orford process is used in extraction of

(a) Fe

(b) Co

(c) Pt

(d) Ni

Answer: D — (d) Ni

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Ni.

293. Horn silver is

(a) AgCl

(b) Ag

(c) AgBr

(d) CH_3COOAg

Answer: A — (a) AgCl

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to AgCl.

294. Which of the following is used in photography

(a) AgCl
(b) AgBr
(c) AgI
(d) Ag_2O

Answer: B — (b) AgBr

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to AgBr.

295. Silver halides are used in photography because

(a) They are photosensitive
(b) Soluble in hypo
(c) Soluble in NH_4OH
(d) Soluble in acids

Answer: A — (a) They are photosensitive

Chemical reason: Silver halides such as AgBr are photosensitive: light decomposes/reduces them to produce tiny silver nuclei that form the latent photographic image.

296. AgCl when heated with Na_2CO_3 gives

(a) Ag_2O
(b) Ag
(c) Ag_2CO_3
(d) $NaAgCO_3$

Answer: B — (b) Ag

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to Ag.

297. $AgNO_3$ gives a red ppt. with

(a) KI
(b) NaBr
(c) $NaNO_3$
(d) K_2CrO_4

Answer: D — (d) K_2CrO_4

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives K_2CrO_4. The choice is consistent with the expected oxidation state and stability of the product.

298. Silver nitrate is prepared by

(a) The action of only conc. HNO_3 on silver
(b) Heating silver oxide with NO_2
(c) The action of hot dil. HNO_3 on silver
(d) Dissolve Ag in aqua–regia

Answer: C — (c) The action of hot dil. HNO_3 on silver

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$. Under the stated conditions this leads to The action of hot dil. HNO_3 on silver.

299. AgCl is soluble in

(a) Aqua–regia
(b) H_2SO_4
(c) HCl
(d) NH_3 (aq)

Answer: D — (d) NH_3 (aq)

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$.

Under the stated conditions this leads to NH_3 (aq).

300. Which of the following is least soluble in water

(a) AgI

(b) AgCl

(c) AgBr

(d) Ag_2S

Answer: D — (d) Ag_2S

Chemical reason: The answer follows from solubility and complex-ion equilibria. Under the stated reagent conditions the equilibrium favours Ag_2S, whereas the alternatives do not show the same dissolution/precipitation behaviour.